

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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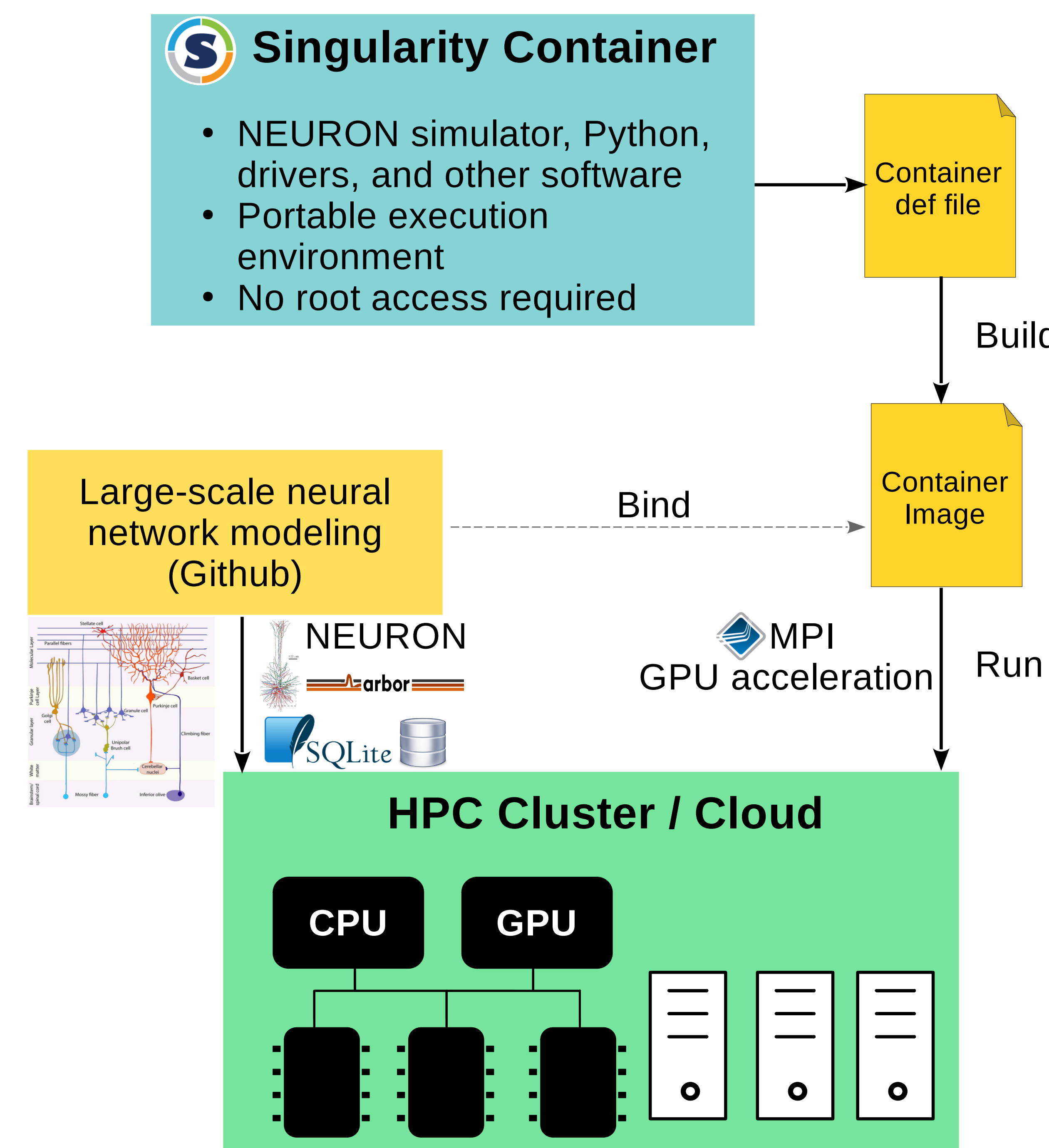
Introduction

1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulate the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

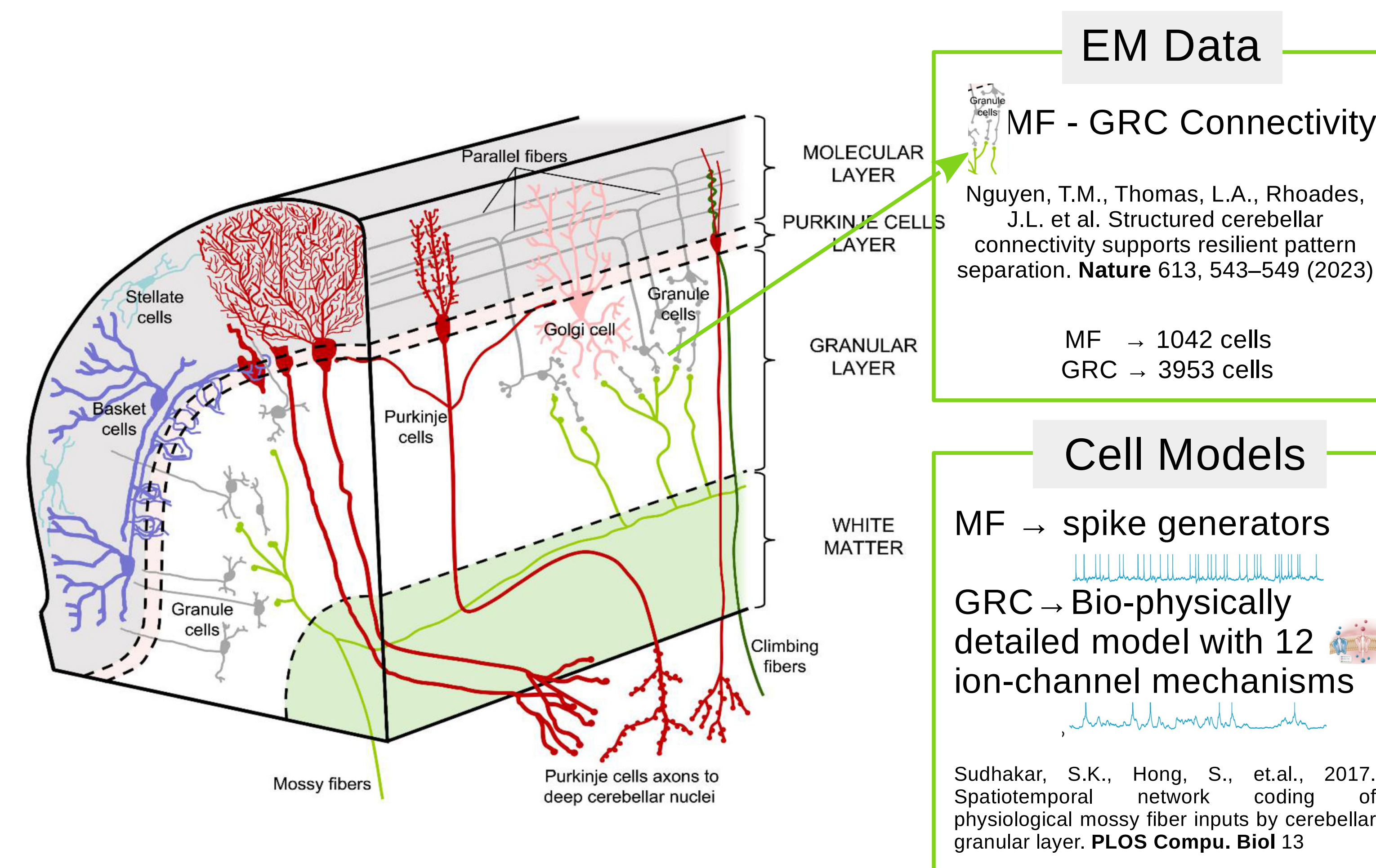
Containerization for Scientific Computing

A **container** is a **lightweight, standalone, virtualized executable environment** that includes all dependencies required to run large-scale scientific models.

Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ($n = 280$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

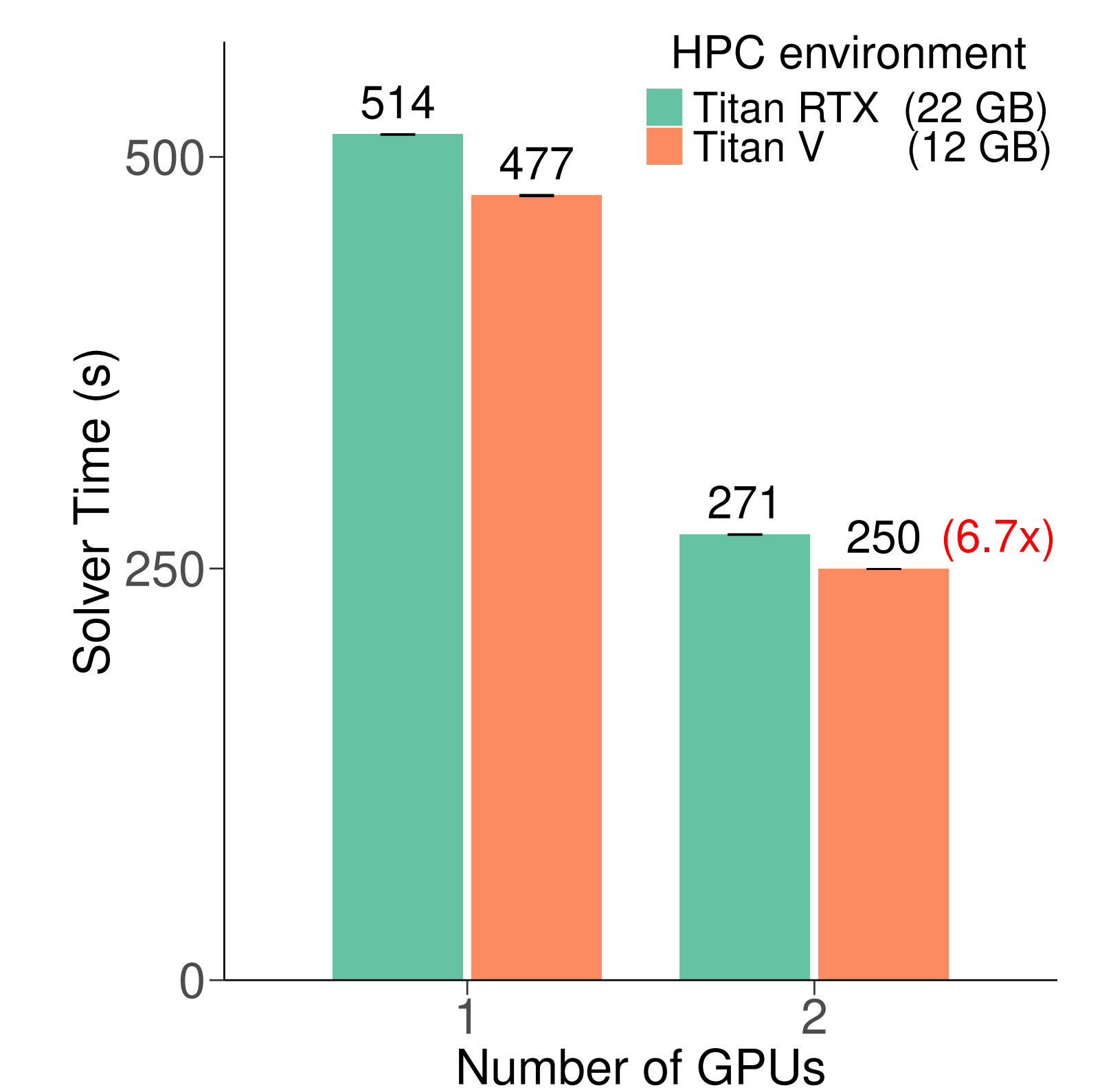
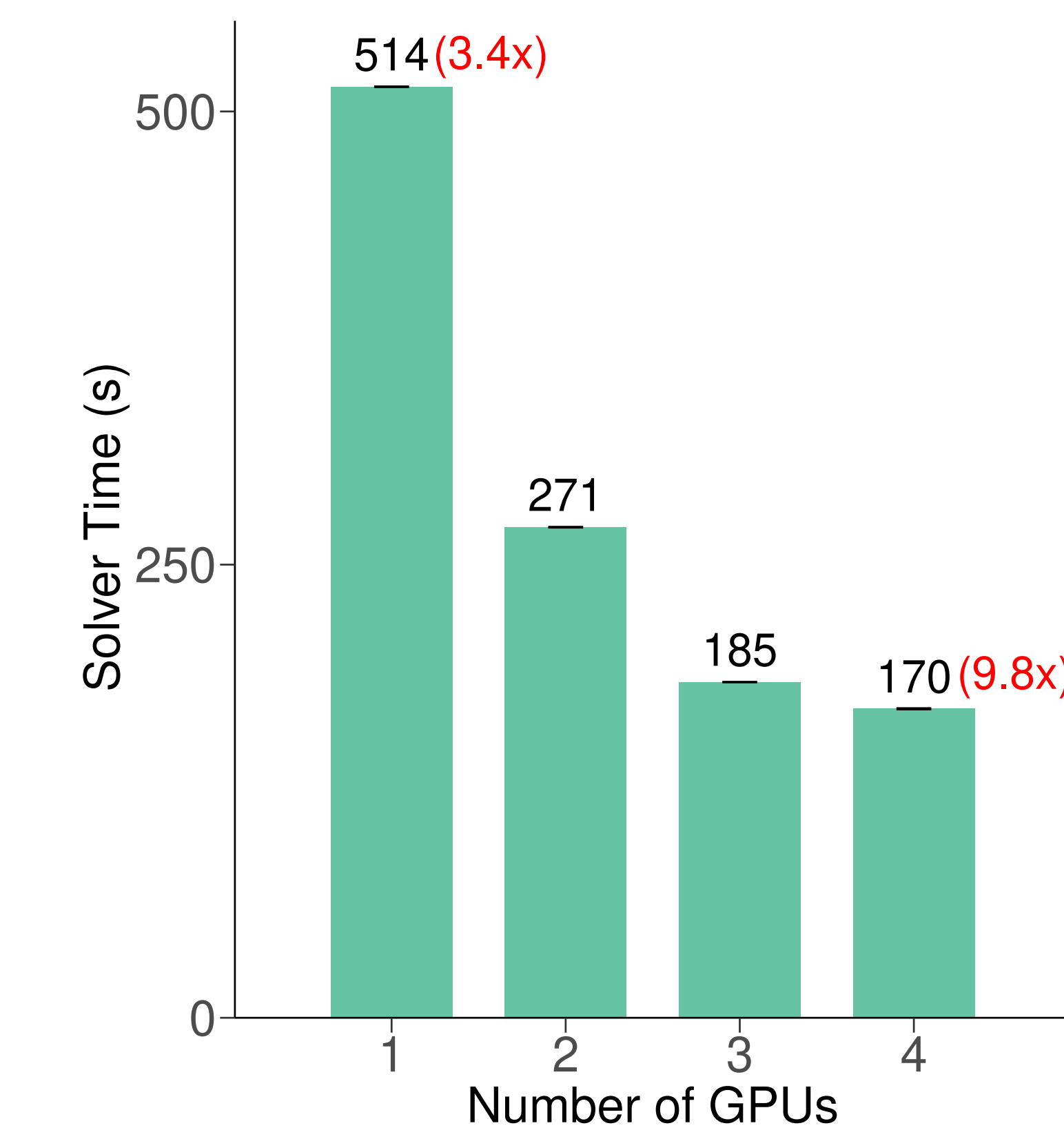
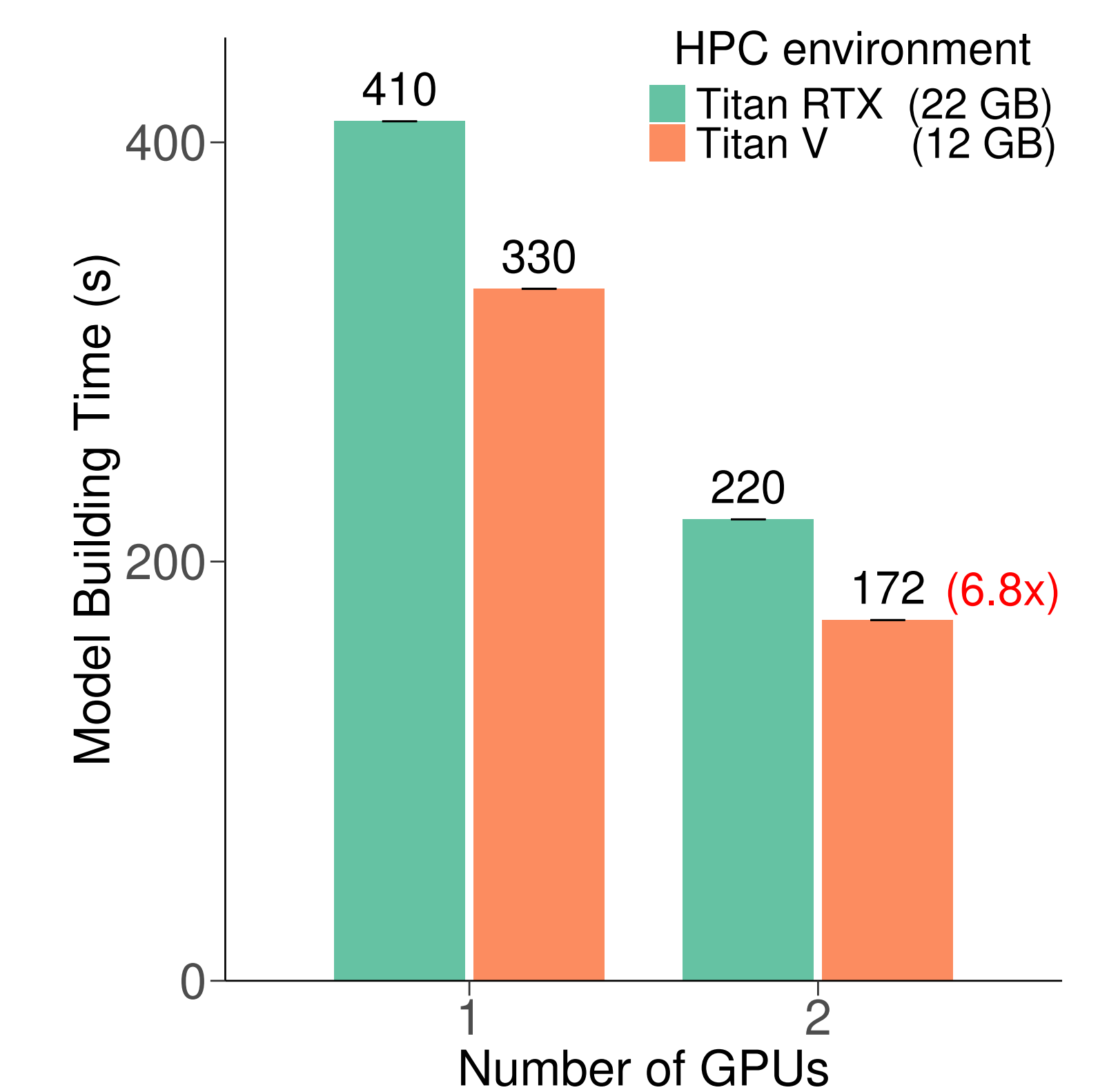
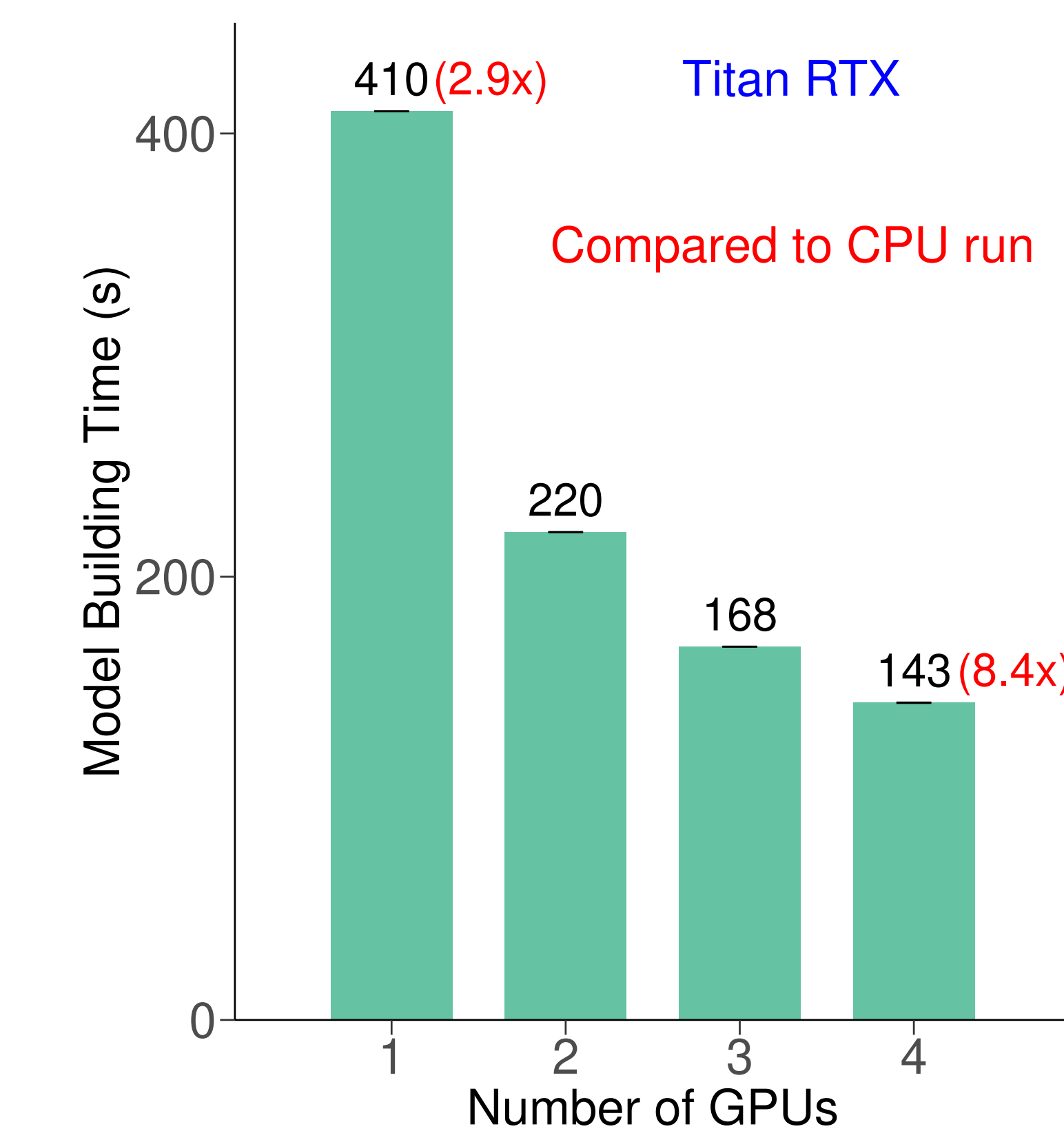
Benchmarking HPC Environments

Table: Network Configuration

Network elements	Count (millions)
MF	0.299
GRC	1.099
Compartments	3.297
Synapses	3.468

Table: Simulation Setup

Simulation time	1 s
Simulation software	CoreNEURON



TITAN RTXs outperform 30-core CPUs by up to 8.4x (on model building) 9.8x (on solver time), with performance improving as GPU count increases.

Future works:

🔧 **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.

🔧 **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

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